

Dummit and Foote Miscellaneous Exercise Solutions

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1 Section 3.4

Problem 8.

Proof. $4 \implies 1$ is obvious.

$1 \implies 2$.

By induction on $|G|$. If G is simple, then $1 \trianglelefteq G$ is a composition series. G is solvable and simple implies G is abelian. ($G/1 \cong G$). By Exercise 1, this implies $G \cong Z_p$ for some prime p . Then $G/1$ is cyclic. If G is not simple, let N be a non-trivial normal subgroup. By Exercise 5, both N and G/N are solvable. By the induction hypothesis, there exists chain of normal subgroups

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_s = N$$

$$N/N = \bar{G}_0 = \bar{N} \trianglelefteq \bar{G}_1 \trianglelefteq \bar{G}_2 \trianglelefteq \dots \trianglelefteq \bar{G}_k = G/N$$

where N_{i+1}/N_i is cyclic for $0 \leq i \leq s-1$ and $G_{i+1}^-/\bar{G}_i$ is cyclic for $0 \leq i \leq k-1$. Then, by the fourth isomorphism theorem, we have

$$\bar{G}_i \trianglelefteq G_{i+1}^- \iff G_i \trianglelefteq G_{i+1}$$

$$G_{i+1}^-/\bar{G}_i \cong G_{i+1}/G_i$$

Therefore, we get

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_s = N \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G_k = G$$

with G_{i+1}/G_i and N_{i+1}/N_i both cyclic, as desired.

$3 \implies 4$.

Let M be the minimal non-trivial normal subgroup of G . If M is simple, then by Exercise 7, G has a composition series one of whose terms is M . By 3, $M/1 \cong M$ is of prime order, therefore $M = Z_p$ for some prime p . Thus M is abelian. If M is not simple, let $N \trianglelefteq M$. Again, by 3, N is of prime index. Say

$$\{N, tN, \dots, t^{p-1}N\}$$

is the partition of M , ie. the set of left cosets of N . Let $x, y \in M$. Then $x \in t^k N, y \in t^l N$ for some $0 \leq k, l < p$. Then,

$$x^{-1}y^{-1}xy \in t^{-k}Nt^{-l}Nt^kNt^lN = N$$

Now, let $g \in G$. Since, $N_G(M) = G$, there exists $a, b \in M$ such that

$$x = gag^{-1}, y = bgg^{-1}$$

Then, we see that

$$x^{-1}y^{-1}xy = ga^{-1}b^{-1}abg^{-1} \in gNg^{-1}$$

Then, $x^{-1}y^{-1}xy$ lies in the intersection of all G -conjugates of N . Let $B = \bigcap_{g \in G} gNg^{-1}$. Since $M \trianglelefteq G$ and $N < M$, $B \leq M$. Also, $B \leq N < M$ so $B \neq M$. Let $x \in B, g \in G$. We know show $gxg^{-1} \in B$. Let $k \in G$. Then, $x \in B \subset g^{-1}kNk^{-1}g$ so

$$x = g^{-1}knk^{-1}g$$

for some $n \in N$. Then,

$$gxg^{-1} = knk^{-1} \in kNk^{-1}$$

Since k was arbitrary, it follows that $gxg^{-1} \in B$ and $B \trianglelefteq G$. As M is the minimal normal subgroup, we conclude $B = 1$ so $x^{-1}y^{-1}xy = 1$ for all $x, y \in M$. Therefore, M is abelian. Now, $|G/M| < |G|$. By induction hypothesis, there exists a chain of subgroups

$$M/M = \bar{G}_0 = \bar{M} \trianglelefteq \bar{G}_1 \trianglelefteq \cdots \trianglelefteq \bar{G}_k = G/M$$

where each $\bar{G}_i$ is a normal subgroup of G/M and $G_{i+1}^{-}/\bar{G}_i$ is abelian. Then, by the fourth isomorphism theorem, we have

$$1 \trianglelefteq M = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$$

where G_{i+1}/G_i is abelian and each G_i is normal in G , as desired.

2 $\implies$ 3.

A finite cyclic group has a normal subgroup of prime order. Using this fact, we can find a normal subgroup G_1^i/H_i of H_{i+1}/H_i . By the Fourth isomorphism theorem, $G_1^i \trianglelefteq H_{i+1}$ and $H_i \trianglelefteq G_1^i$. Continue extracting subgroups $G_2^i, G_3^i, \dots$ in a similar fashion until G_{j+1}^i/G_j^i is simple. Then, we get a composition series of G where each composition factor is of prime order. $\square$

Problem 4.

Proof. Let G be a finite abelian group. Let $n||G|$ and $p|n, p$ prime. By Cauchy's Theorem, there exists an element $x \in G$ such that $| \langle x \rangle | = p$. Since, G is

abelian $\langle x \rangle = H$ is normal in G . G/H is of order $|G|/p$. By the induction hypothesis, G/H has a subgroup M/H of order $\frac{n}{p}$. Then,

$$B = \bigcup_{m \in M} mH$$

is a subgroup of G of order $\frac{n}{p} \cdot p = n$ □

2 Section 3.5

Problem 4.

Proof. Let $a = (123 \dots n), b = (12)$. Notice that

$$(1i) = (a^{i-1}(ba)^{n-2i+2})^{n-2}$$

□

Section 4.5

1) Sylow-p subgroups exist.

by induction on $|G|$. Let $|G| = p^m k$. For $m = 1$, Cauchy theorem provides us with an element of order p . If $p \nmid Z(G)$, then writing the class equation

$$|G| = |Z(G)| + \sum_{i=1}^n [G : C_G(g_i)]$$

where g_i are representatives of non-central conjugacy classes. If $p \nmid [G : C_G(g_i)]$ for all i , then $p \mid |Z(G)|$. Therefore, there is g_j such that

$$p \nmid [G : C_G(g_j)] \implies |C_G(g_j)| = p^m h$$

for some $h \mid k$. By the induction hypothesis, there exists a Sylow-p subgroup of H . This subgroup is also a Sylow-p subgroup of G . If $p \mid Z(G)$, then by Cauchy's theorem, there exists $x \in Z(G)$ with $|x| = p$. Being a subgroup of $Z(G)$, $\langle x \rangle = N$ is normal in G and we have $|G/N| = p^{m-1}k$. By the induction hypothesis, there exist Sylow-p subgroup P' of $\overline{G}$. Take the corresponding subgroup of G , then we have $|P| = |P'/N| \cdot |N| = p^{m-1} \cdot p = p^m$. □

4.5-17 n_5 is either 1 or 21. Similarly, n_7 is either 1 or 15. Suppose $n_5 = 21$. Then there are $21 \cdot 4 = 84$ elements of degree 5 in G . In this case, $n_7 = 1$ for otherwise we have $84 + 6 \cdot 15 > 105$ elements in the group. Let $Q \in \text{Syl}_7(G)$. Since $n_7 = 1$, $Q \trianglelefteq G$ and $|G/Q| = 15$. Then G/Q has a normal Sylow 5-subgroup $\overline{P}$. The subgroup P of G corresponding to $\overline{P} \in G/Q$ has degree $|\overline{P}| \cdot |Q| = 35$. Since $n_7 = 1$ and $n_5(G/Q) = 1$, P must be the only subgroup of degree 35

in G . However, for $H, H' \in \text{Syl}_5(G)$, since $H, H' \cong Z_5$ and $Q \cong Z_7$, we have $|\langle H, Q \rangle| = |\langle H', Q \rangle| = 35$. Since $H \cap H' = \{1\}$ for $H, H' \in \text{Syl}_5(G)$, it follows that there are 21 subgroups of G of order 35! $\Rightarrow \Leftarrow$. Thus, $n_5 = 1$.

Now, assume $n_7 = 21$. Then, $n_5 = 1$. Let $P \in \text{Syl}_5(G)$ then $P \trianglelefteq G$ and $n_7(G/P) = 1$. By the same logic as before, there exists only one subgroup of order 35 in G . However, for $H, H' \in \text{Syl}_7(G)$, we have $\langle H, P \rangle \neq \langle H', P \rangle$ and $|\langle H, P \rangle| = |\langle H', P \rangle| = 35$ Contradiction! Thus, $n_7 = 1$ as well.

5.4-9 If P is cyclic, nothing to show. Assume P is not cyclic. Then, $P/Z(P)$ must have order p^2 . (If it has order p , then the quotient is cyclic and so is P .) Now, if $P/Z(P)$ is cyclic of order p^2 then, P is cyclic and nothing to show. So assume $P/Z(P)$ is not cyclic. Then, since there is no nontrivial homomorphism from Z_p to $\text{Aut}(Z_p)$, we must have

$$P/Z(P) \cong Z_p \times Z_p \cong \langle a \rangle \times \langle b \rangle$$

Now, let $x \in P$. Then, there are $k, l \leq p-1$ such that

$$x \in a^k b^l Z(P)$$

and

$$\phi(x) = x^p \in (a^k b^l)^p Z(P) = Z(P)$$

So ϕ is indeed a map from P to $Z(P)$. It is now easy to check that ϕ is also a homomorphism.

4.34 Let $Q \in \text{Syl}_p(N)$ and $P \in \text{Syl}_p(G)$. Then, there exists $g \in G$ with $gQg^{-1} \subset P$. It follows that $N \cap P \subset gQg^{-1} \subset P$ and therefore $gQg^{-1} \subset N \cap P$. This shows that $N \cap P \in \text{Syl}_p(N)$

5.5-6 Assume $K \cong Z_r = \langle k \rangle$. Then, there is $\sigma \in \text{Aut}(Z_r)$ such that $\sigma\psi_1(k)\sigma^{-1} = \psi_2(k^a) = \psi_2(k)^a$. It follows that $\sigma\psi_1(t)\sigma^{-1} = \psi_2(t^a) = \psi_2(t)^a$ for all $t \in K$. Since ψ_1 and ψ_2 are conjugate, the map

$$\phi : k \rightarrow k^a$$

must be an automorphism of $K \cong Z_r$. It follows that $(a, r) = 1$. Now, it is easy to show that

$$\begin{aligned} \rho : H \rtimes_{\psi_1} K &\rightarrow H \rtimes_{\psi_2} K \\ (h, k) &\mapsto (\sigma(h), k^a) \end{aligned}$$

is an isomorphism.

5.5-5b Write $H = \langle a \rangle \times \langle b \rangle$ and $K = S_3 \cong \text{Aut}(H)$ and $G = H \rtimes_{\phi} K$ where ϕ is the identity map from K to $\text{Aut}(H)$. As suggested in the hint, let $\psi : G \rightarrow S_4$ be the homomorphism obtained by letting G act on left cosets of K . The left cosets of K are

$$K, aK, bK, abK$$

and for $g \in G$,

$$\psi(g)(mK) = \sigma_g(mK) = gmK$$

Then, we see that

$$\ker \psi = \bigcap_{g \in G} gKg^{-1}$$

In particular, $\ker \psi \leq K \cap aKa^{-1}$. If $k' \in K \cap aKa^{-1}$ then, there is $k \in K$ such that

$$aka^{-1} = k' \implies aka^{-1}k^{-1} = k'k^{-1}$$

Since $H \cap K = \{1\}$, we see that $k'k = 1$ and $aka^{-1}k^{-1} = 1 \implies ka^{-1}k^{-1} = a^{-1}$. It follows that if $k \in \ker \psi$, then k is mapped to the identity automorphism on H . ie

$$k \in \ker \psi \implies k = \phi(k) = 1 \iff k \in \ker \phi = 1$$

Thus, the action of G on the left cosets of K is transitive and we see that $\ker \psi = 1$ so ψ is an isomorphism. Hence, $G \cong S_4$.

5.5-9 $1805 = 5 \cdot 19^2$. Let $|G| = 1805$. It follows that $n_{19} = 1$. Let P be the Sylow-19 subgroup of G . Let A be the given matrix. Then, $G = P \rtimes \langle A \rangle$. If $P = Z_{19^2}$, then $\text{Aut}(P) \cong Z_{18 \cdot 19}$ and the only homomorphism from A to $\text{Aut}(P)$ is the trivial homomorphism. In this case $G = Z_{19^2} \times Z_5$ is abelian. If $P = Z_{19} \times Z_{19}$, then $\text{Aut}(P) \cong GL_2(\mathbb{F}_{19})$ and is of order $19^2(19^2 - 1) = 5 \cdot 2^4 \cdot 3^4 \cdot 19$. Since all Sylow 5 subgroups of this group are isomorphic, all non-trivial homomorphisms from Z_5 to P give rise to isomorphic semidirect products $P \rtimes \langle A \rangle$. Thus, there is a unique non-abelian group of order 1805 and also two abelian groups of order 1805, namely $Z_{19^2} \times Z_5$ and $Z_{19} \times Z_{19} \times Z_5$.

Section 7.5

7.5-3 We consider the homomorphism φ given in section 3 exercise 26. Assume F is infinite, ie F has characteristic 0. Then, $\mathbb{Z} \cong \varphi(\mathbb{Z}) \leq F$. So by Corollary 16, the subfield generated by $\varphi(\mathbb{Z})$ is isomorphic to $\mathbb{Q}$, the field of fractions of $\mathbb{Z}$. Now, we claim the unique smallest field F_0 is $\varphi(\mathbb{Z})$. Let $k \in \varphi(\mathbb{Z})$, then $k = \varphi(t) = 1 + \dots + 1$. Since $1 \in F_0$, it follows that $k \in F_0$. This proves $F_0 = \varphi(\mathbb{Z})$. If F is finite, then its characteristic is a prime number p . Then, $\ker \varphi = p\mathbb{Z}$. Then, we get $\mathbb{Z}/p\mathbb{Z} \cong \varphi(\mathbb{Z}) \leq F$. Again, by corollary 16, the subfield of F generated by $\varphi(\mathbb{Z})$ is isomorphic to the field of fractions of $\mathbb{Z}/p\mathbb{Z}$, which is $\mathbb{Z}/p\mathbb{Z}$ itself.

7.5-4 Let F_0 be the unique smallest subfield of $\mathbb{R}$. By the previous exercise, we know $F_0 \cong \mathbb{Q}$. Since $\mathbb{Q} \leq \mathbb{R}$, it follows that $F_0 = \mathbb{Q}$.

Section 7.3

7.4-25 By induction on n . Simply notice that in the expression

$$\sum \binom{n}{k} a^{k+1} b^{n-k} + \sum \binom{n}{k} a^k b^{n-k+1}$$

the term $a^{j+1} b^{n-k}$ has factor

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$$

Section 7.4

7.4-35 Let P be the set of all ideals I not containing A . Order P by inclusion. Let $\mathcal{C}$ be a chain in P . ie $\mathcal{C} = \{I_n\}_{n \in \mathbb{N}}$ where $I_k \leq I_{k+1}$ for all $k \geq 1$. Then, by earlier exercises, $\bigcup_n I_n$ is an ideal. Then by Zorn's Lemma, P has a maximal element.

7.6

7.6-1 It is easy to check Re and $R(1-e)$ are ideals. Define the map $\varphi : R \rightarrow Re \times R(1-e)$ by

$$\varphi(r) = (re, r(1-e))$$

It is easy to check that $(1-e)^2 = (1-e)$. Then it follows that $e, (1-e)$ are the identities of $Re, R(1-e)$ respectively. Then we see that if $r \in \ker \varphi$, then $re = 0, r(1-e) = 0 \implies r - re = r - 0 = 0 \implies r = 0$. Then by the second isomorphism theorem for rings, we conclude $R \cong Re \times R(1-e)$.

Alfonzo's Notes Exercise 7 (Lemma 17) In a UFD, all irreducibles are prime.

Proof. Let R be a UFD. Let $p \in R$ be an irreducible. Assume $p|ab$ for some $a, b \in R$. Since R is a UFD, there exists $q_i, r_i \in R$ irreducible and $k \in R$ such that $ab = pk$ and

$$q_1 \dots q_n \cdot r_1 \dots r_m = pk$$

where $\prod q_i = a, \prod r_j = b$. Again, by UFD property of R , there exists irreducibles k_j with $k = \prod k_j$. Then,

$$q_1 \dots q_n \cdot r_1 \dots r_m = p \cdot k_1 \dots k_{n+m-1}$$

By the uniqueness of factorization into irreducibles, we can assume q_1 and p are associates. That means $p|q_1$ and so $p|a$. $\square$

8.3-6

- a) The elements of $\mathbb{Z}[i]/(i+1)$ are $(i+1), i+(i+1)$ as $a+bi = a(i+1) + (b-a)i$ and $\gcd(i+1, i) = 1$ is easily seen by considering the norms.
- b) In $\mathbb{Z}[i]$, the prime p is irreducible if $p \equiv 3 \pmod{4}$. Then, we see that (p) is maximal and so $\mathbb{Z}[i]/(p)$ is a field. The elements of this field are of the form $a+bi$ where $a, b \in \mathbb{Z}/p\mathbb{Z}$ and therefore the order of this field is p^2 .
- c) If $p = \pi\bar{\pi}$ is given, we need to check that $(\pi), (\bar{\pi})$ are comaximal to conclude the claim. However, this follows easily since in $\mathbb{Z}[i]$, the irreducible factors of p are π and $\bar{\pi}$ and so there exists $x, y \in \mathbb{Z}[i]$ such that $x\pi + y\bar{\pi} = 1$, implying $(\pi) + (\bar{\pi}) = \mathbb{Z}[i]$.
- 8.3-7** a) $a + \pi k \mapsto a\pi^n + \pi^{n+1}k \in (\pi^n)/(\pi^{n+1})$. And $a\pi^n + \pi^{n+1}k \in (\pi^{n+1}) \iff a \in (\pi)$ so the map is one-to-one. It is easy to see that multiplication by π^n is surjective. Hence, $\mathbb{Z}[i]/(\pi) \cong (\pi^n)/(\pi^{n+1})$.
- b) COMING SOON.

c) Since $\mathbb{Z}[i]$ is a UFD, given $\alpha \in \mathbb{Z}[i]$ we can write

$$\alpha = (1+i)^{\alpha_1} p^{\alpha_2} (a+bi)^{\alpha_3} (a-bi)^{\alpha_4}$$

where $p \equiv 3 \pmod{4}$ and $a^2 + b^2 = q \equiv 1 \pmod{4}$.

8.3-8 b) We proceed as the hint suggests. $R/I_3 \cong R/(3)/I_3/(3)$. $R/(3)$ is a set of the form $a + b\sqrt{-5}$ where $a, b \in \mathbb{Z}_3$. So there are 9 elements in this ring. Now consider $\alpha = 3r + (2 + \sqrt{-5})l = 3r + 2l + \sqrt{-5}l$ if $l \equiv 1 \pmod{3}$ then $\alpha \in 1 + 2\sqrt{-5} + (3)$, if $l \equiv 2 \pmod{3}$ then $\alpha \in 2 + \sqrt{-5} + (3)$ and if $l \equiv 0 \pmod{3}$ then $\alpha \in (3)$. This shows the elements of $I_3/(3)$ are $(3), 2 + \sqrt{-5} + (3)$ and $1 + 2\sqrt{-5} + (3)$. So the latter quotient ring has 3 elements. Thus, $|R/I_3| = 9/3 = 3 \implies R/I_3 \cong \mathbb{Z}/3\mathbb{Z}$ which shows I_3 is maximal and thus, prime. Similar argument works for I_3' and I_2 as well.

Section 9.3

9.3-1 Let $p(x) = a(x)b(x)$ with $p(x) \in R[x], a(x), b(x) \in F[x]$ all monics. Multiply by a common denominator $de \in R$ to lift the coefficients of $a(x), b(x)$ from F to R .

$$edp(x) = (ex^n + e(\dots))(dx^m + d(\dots))$$

Now, assume R is a UFD. Then $R[x]$ is a UFD. Then, as in the proof of Gauss's lemma, we can distribute the divisors of ed to $ea(x)$ and $db(x)$ in a way that the factors stay in $R[x]$. Assume we can divide both sides by ed like this: $ed = pq$ and $p|a(x), q|b(x)$. Then,

$$p(x) = \left(\frac{e}{p}x^n + \frac{e}{p}(\dots)\right)\left(\frac{d}{q}x^m + \frac{d}{q}(\dots)\right) = a'(x)b'(x)$$

Now, $a'(x), b'(x) \in R[x]$. Since $p(x)$ is monic, we must have both e/p and d/q units in $R[x]$. Then, we can multiply $a'(x)$ by d/q , $b'(x)$ by e/p and $p(x)$ by

$ed/pq = 1$ to see that

$$\frac{d}{q} \left(\frac{e}{p} x^n + \frac{e}{p} (\dots) \right) = (x^n + \dots) = a(x) \in R[x]$$

9.3-2 Write

$$f(x)g(x) = (a_n x^n + \dots + a_1 x + a_0)(b_m x^m + \dots + b_1 x + b_0) = p(x) \in \mathbb{Z}[x]$$

Multiply by a common denominator to lift the coefficients of $f(x), g(x)$ to $\mathbb{Z}$

$$(ef(x))(dg(x)) = edp(x)$$

Assume ed distributes its factors to $ef(x)$ and $dg(x)$ as $ed = pq$, and $p|ef(x), q|dg(x)$. Then $\frac{e}{p}f(x), \frac{d}{q}g(x) \in \mathbb{Z}[x]$ and

$$\frac{e}{p}f(x) \cdot \frac{d}{q}g(x) = p(x)$$

Notice that the product of coefficients of these functions is an integer. Then it follows that

$$a_r \cdot b_s = \frac{ed}{pq} \cdot a_r \cdot b_s = \left(\frac{e}{p}a_r\right)\left(\frac{d}{q}b_s\right) \in \mathbb{Z}$$

as desired.